Qualitative Error Analysis: Research!!

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1. What Is Qualitative Error Analysis?

- Qualitative error analysis involves examining individual system outputs to understand why errors occur, rather than simply measuring how many errors occur.
- It focuses on identifying patterns, recurring weaknesses, and the underlying causes of incorrect or unreliable results.
- Unlike quantitative evaluation, which relies on numerical metrics, qualitative analysis involves manually reviewing outputs and comparing them against expected results and supporting evidence.
- The goal is to understand not only *what* went wrong, but also *why* it happened and whether similar errors appear across multiple evaluations.

Why it matters:

- Identifies weaknesses that overall accuracy scores might overlook.
- Helps distinguish between different types of system failures.
- Reveals whether errors originate from missing information, incorrect interpretation, unsupported conclusions, or other issues.
- Provides context for understanding the reliability of AI-generated assessments.

Source: ARES: An Automated Evaluation Framework for Retrieval-Augmented Generation Systems (Saad-Falcon et al., 2024)

2. Common Types of Errors to Investigate

Research on Retrieval-Augmented Generation (RAG) systems identifies several categories of errors that can occur when retrieving information, interpreting source material, and generating conclusions.

The following categories provide a potential framework for organizing qualitative observations.

A. Retrieval Errors

Errors involving the information selected or retrieved from the source material.

- **Missing retrieval:** Relevant information exists but is not retrieved.
- **Incomplete retrieval:** Only part of the necessary information is retrieved.
- **Irrelevant retrieval:** The system retrieves information that is related to the topic but does not directly support the requirement.
- **Context fragmentation:** Information is retrieved without enough surrounding context to understand its full meaning.

B. Interpretation Errors

Errors involving how the system understands retrieved information.

- **Misinterpretation:** Incorrectly understanding the meaning of a technical requirement.
- **Context loss:** Overlooking important conditions, exceptions, or qualifications.
- **Overgeneralization:** Applying a statement more broadly than the original source allows.
- **Terminology confusion:** Misunderstanding technical language or treating different concepts as interchangeable.

C. Evidence and Citation Errors

Errors involving the relationship between the system's conclusions and its supporting evidence.

- **Unsupported claims:** Making conclusions that cannot be verified using the available documentation.
- **Incorrect citations:** Referencing a source that does not support the associated conclusion.
- **Evidence misalignment:** Providing relevant information that does not actually demonstrate that a specific requirement is satisfied.
- **Hallucination:** Generating information that is not present in, or is contradicted by, the retrieved source material.

D. Classification Errors

Errors involving the system's final assessment.

- **False positive:** Incorrectly identifying a requirement as compliant when sufficient supporting evidence is not available.
- **False negative:** Incorrectly identifying a requirement as noncompliant when sufficient supporting evidence exists.
- **Incorrect confidence:** Presenting an assessment with more certainty than the available evidence justifies.

Relevant research:

1. Classifying and Addressing the Diversity of Errors in Retrieval-Augmented Generation Systems (Leung et al., 2026) — Develops a taxonomy of RAG errors and discusses how different failure types can be identified and addressed.
2. RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models (Niu et al., 2024) — Studies unsupported and contradictory information in RAG-generated responses, including human annotations identifying hallucinations at the individual text-span level.

3. How to Conduct a Qualitative Error Analysis

Based on existing research, a structured manual review process can help ensure that errors are consistently identified and documented.

Step 1: Establish an Expected Result

- Identify what the correct result should be for each software requirement being tested.
- Use NASA's original documentation to understand the rules and what each requirement needs to follow.
- Determine what evidence is needed to decide whether a requirement is compliant or non-compliant.
- Compare the AI's response with the expected result to check whether the system made the correct decision.

Step 2: Review Individual Outputs

- Examine the system's generated assessment for each requirement.

- Compare its conclusion against the original requirement and supporting documentation.
- Verify that cited evidence actually supports the conclusion.
- Identify missing information, incorrect interpretations, or unsupported claims.
- Look beyond whether the final answer is correct and examine how the system arrived at it.

Step 3: Categorize Identified Errors

- Assign each error to an appropriate category (retrieval, interpretation, evidence, or classification).
- Record the specific issue observed.
- If an error falls into multiple categories, document the contributing factors.
- Avoid treating every incorrect output as the same type of failure.

Step 4: Investigate Potential Root Causes

- Determine what may have contributed to the error.
- Examine whether the issue originated from missing evidence, ambiguous wording, incorrect interpretation, or another identifiable limitation.
- Distinguish between an observed error and a hypothesized underlying cause.
- Look for similar errors across different requirements to determine whether a recurring pattern exists.

Step 5: Identify Recurring Patterns

- Group similar errors together.
- Determine which error categories appear most frequently.
- Identify whether particular types of requirements are more difficult to evaluate accurately.
- Examine whether multiple errors share a common underlying cause.
- Document examples that demonstrate particularly important or recurring weaknesses.

Step 6: Document Findings

- Summarize the most significant error categories.

- Include representative examples of incorrect outputs.
- Explain why each example constitutes an error.
- Record the potential cause and its implications for system reliability.
- Document cases where the system produced correct results as well, providing context for its strengths and limitations.

4. Recommended Documentation Structure

A useful approach is to create an error analysis table that records each reviewed case in a consistent format.

Field	Description
Requirement ID	Identifier of the requirement being evaluated
Expected Result	Correct assessment based on the source documentation
Actual Result	Assessment generated by the system
Error Category	Retrieval, interpretation, evidence, or classification
Observed Issue	Description of what specifically went wrong
Supporting Evidence	Relevant source clause or documentation
Potential Root Cause	Explanation of why the error may have occurred
Severity	Assessment of how significantly the error affects reliability

Example (illustrative):

Field	Example
Requirement ID	REQ-012

Expected Result	Noncompliant
Actual Result	Compliant
Error Category	Evidence / Classification
Observed Issue	The system identified a general statement as sufficient evidence, although the specific requirement was not demonstrated.
Supporting Evidence	The cited clause does not establish the required condition.
Potential Root Cause	The system may have interpreted related information as direct evidence of compliance.
Severity	High

Note: This is a hypothetical example to demonstrate how an error could be documented, not an observed result.

5. Important Considerations When Analyzing Errors

Human Review and Consistency

- Human review is valuable because automated evaluation metrics may not capture the full context of an incorrect output.
- Having two reviewers independently examine the same outputs can help identify differences in interpretation.
- Reviewers should establish consistent definitions for each error category before beginning the analysis.
- Disagreements should be documented and resolved through discussion using the original source material as the reference.

Analyzing Correct Outputs

- Qualitative analysis should not exclusively examine incorrect results.

- Reviewing correctly classified examples helps establish what the system handles successfully.
- Comparing correct and incorrect cases can reveal conditions under which the system performs reliably or struggles.
- This also helps prevent drawing conclusions based only on isolated failures.

Distinguishing Symptoms From Root Causes

- An incorrect compliance assessment is an observable outcome, not necessarily an explanation of why the error occurred.
- For example, a false positive might be caused by incomplete retrieval, incorrect interpretation, or insufficient evidence evaluation.
- Root-cause analysis should therefore examine the evidence and reasoning associated with each individual failure.
- Avoid assuming a cause unless the available evidence supports it.

Severity and Impact

Not all errors have the same implications.

A possible severity classification is:

- **Low:** Minor inaccuracies that do not materially change the overall assessment.
- **Medium:** Incomplete or misleading information that affects the explanation or supporting evidence.
- **High:** Errors that directly change the final compliance classification or result in an unsupported conclusion.

Severity should be determined based on the potential impact of the error, rather than simply how frequently it occurs.

6. Existing Evaluation Frameworks and Their Relevance

Several established research frameworks provide useful concepts for evaluating RAG systems.

RAGAS (Es et al., 2024)

Research Paper

- Introduces a framework for evaluating RAG systems across different dimensions.
- Includes context precision, context recall, faithfulness, and answer relevance.
- Separates the quality of retrieved information from the quality of generated responses.
- Particularly useful as a reference for understanding how retrieval and generation errors can be examined independently.

ARES (Saad-Falcon et al., 2024)

Research Paper

- Evaluates RAG systems using context relevance, answer faithfulness, and answer relevance.
- Incorporates human-annotated examples to improve the reliability of automated evaluation.
- Demonstrates the value of combining automated assessments with human judgment.
- Useful for understanding why manual evaluation remains important even when automated metrics are available.

RAGTruth (Niu et al., 2024)

Research Paper

- Provides a human-annotated dataset of approximately 18,000 RAG-generated responses.
- Identifies hallucinations at both the response and individual text-span levels.
- Distinguishes between unsupported and contradictory information.
- Demonstrates how individual errors can be annotated systematically rather than simply labeling an entire response as incorrect.

7. Key Takeaways

- Qualitative error analysis should focus on understanding the underlying reasons behind incorrect system outputs, not just counting failures.

- Errors should be separated into distinct categories to identify where and how failures occur.
- Every identified error should be compared against the original technical documentation and its supporting evidence.
- Root-cause analysis is important for distinguishing between an incorrect final assessment and the underlying issue that produced it.
- Reviewing both correct and incorrect outputs provides a more balanced understanding of system reliability.
- Consistent documentation makes it easier to identify recurring patterns and communicate findings.
- Human judgment is especially important when evaluating technical language, evidence sufficiency, and nuanced compliance classifications.

Overall conclusion: A successful qualitative error analysis should produce more than a list of incorrect outputs. It should provide a structured explanation of the system's recurring weaknesses, the circumstances in which those weaknesses occur, and the extent to which they affect the reliability of its assessments.

8. References

1. Leung, K. K., et al. (2026). *Classifying and Addressing the Diversity of Errors in Retrieval-Augmented Generation Systems.* EACL 2026. https://aclanthology.org/2026.eacl-long.147/
2. Niu, C., et al. (2024). *RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models.* ACL 2024. https://aclanthology.org/2024.acl-long.585/
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